Chapter 3, Section 11

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1. Show that the result of (11.2) is false without the projective hypothesis. For example, let $X = \mathbb{A}^n_k$, let $P = (0, \ldots, 0)$, let $U = X - P$, and let $f : U \to X$ be the inclusion. Then the fibers of f all have dimension 0, but $\mathrm{R}^{n-1} f_* \mathcal{O}_U \neq 0$.

Proof. By (8.2) and (8.5), we have the isomorphism

$$\mathrm{R}^{n-1} f_* \mathcal{O}_U \big|_U \cong \widetilde{\mathrm{H}^{n-1}(U, \mathcal{O}_U)},$$

so it suffices to show $\mathrm{H}^{n-1}(U, \mathcal{O}_U) \neq 0$. This is a generalization of (Ex. 4.3), and we compute $\mathrm{H}^{n-1}(U, \mathcal{O}_U)$ via Čech cohomology. Let $x_1, \ldots, x_n$ be affine coordinates of X, i.e., $X = \operatorname{Spec} k[x_1, \ldots, x_n]$, and take $\mathfrak{U}$ to be the open covering of U by the open sets $U_i = X - \{x_i \neq 0\}$. Setting $V = \bigcap_{i=1}^n U_i$ and $V_j = \bigcap_{i \neq j} U_i$, the last two terms of the Čech complex are

$$C^{n-2} = \prod_{j=1}^n \Gamma(V_j, \mathcal{O}_U), \quad C^{n-1} = \Gamma(V, \mathcal{O}_U).$$

Now

$$\Gamma(V_j, \Omega) = k\left[x_1, \ldots, x_n, \frac{1}{x_1}, \ldots, \widehat{\frac{1}{x_j}}, \ldots, \frac{1}{x_n}\right],$$
$$\Gamma(V, \Omega) = k\left[x_1, \ldots, x_n, \frac{1}{x_1}, \ldots, \frac{1}{x_n}\right],$$

and the image of the boundary map $d : C^{n-2} \to C^{n-1}$ consists of all expressions $x_1^{r_1} \cdots x_n^{r_n}$ where at least one of $r_1, \ldots, r_n$ is non-negative. Hence, $\mathrm{H}^{n-1}(U, \mathcal{O}_U)$ consists of all expressions $x_1^{r_1} \cdots x_n^{r_n}$ with $r_i < 0$ for all i. □

2. Show that a projective morphism with finite fibers is a finite morphism.

Proof. Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $Y' = \mathcal{Spec}\, f_* \mathcal{O}_X$. The natural map $g : Y' \to Y$ is finite, and we get a projective morphism $f' : X \to Y'$ with connected fibres such that f factors into $g \circ f'$ (11.5). The property of quasi-finiteness satisfies the conditions of (II, Ex. 4.8), and finite morphisms are separated, so f' is quasi-finite. The composition of finite morphisms is finite, so it suffices to show f' is a finite morphism. Thus, we reduce to showing projective morphisms with finite and connected fibres is a finite morphism. By (II, Ex. 3.4), we further reduce to the case $Y = \operatorname{Spec} A$ for some Noetherian ring A and show $X = \operatorname{Spec} B$, where B is a finite A-algebra. If f has finite and connected fibres, then f is injective as a map between topological spaces. Since projective morphisms are closed mappings, f is a homeomorphism of X onto a closed subscheme of Y. The result follows from (II, Ex. 3.11). □

3. Let X be a normal, projective variety over an algebraically closed field k. Let $\mathfrak{d}$ be a linear system (of effective Cartier divisors) without base points, and assume that $\mathfrak{d}$ is *not composite with a pencil*, which means that if $f : X \to \mathbb{P}^n_k$ is the morphism determined by $\mathfrak{d}$, then $\dim f(X) \geq 2$. Then show that every divisor in $\mathfrak{d}$ is connected. This improves Bertini's theorem (10.9.1).

4. *Principle of Connectedness.* Let $\{X_t\}$ be a flat family of closed subschemes of $\mathbb{P}^n_k$ parameterized by an irreducible curve T of finite type over k. Suppose there is a nonempty open set $Y \subseteq T$, such that for all closed points $t \in Y$, X_t is connected. Then prove that X_t is connected for all $t \in T$.

5. Let Y be a hypersurface in $X = \mathbb{P}_k^N$ with $N \geq 4$. Let $\hat{X}$ be the formal completion of X along Y (II, §9). Prove that the natural map $\operatorname{Pic} \hat{X} \to \operatorname{Pic} Y$ is an isomorphism.

6. Again let T be a hypersurface in $X = \mathbb{P}_k^N$, this time with $N \geq 2$.

(a) If $\mathcal{F}$ is a locally free sheaf on X, show that the natural map

$$H^0(X, \mathcal{F}) \to H^0(\hat{X}, \hat{\mathcal{F}})$$

is an isomorphism.

(b) Show that the following conditions are equivalent:

(i) for each locally free sheaf $\mathfrak{F}$ on $\hat{X}$, there exists a coherent sheaf $\mathcal{F}$ on X such that $\mathfrak{F} \cong \hat{\mathcal{F}}$ (i.e., $\mathfrak{F}$ is *algebraizable*);

(ii) for each locally free sheaf $\mathfrak{F}$ on $\hat{X}$, there is an integer n_0 such that $\mathfrak{F}(n)$ is generated by global sections for all $n \geq n_0$.

(c) Show that the conditions (i) and (ii) of (b) imply that the natural map $\operatorname{Pic} X \to \operatorname{Pic} \hat{X}$ is an isomorphism.

Note. In fact, (i) and (ii) always hold if $N \geq 3$. This fact, coupled with (Ex. 11.5) leads to Grothendieck's proof of the Lefschetz theorem which says that if Y is a hypersurface in $\mathbb{P}_k^N$ with $N \geq 4$, then $\operatorname{Pic} Y \cong \mathbb{Z}$, and it is generated by $\mathcal{O}_Y(1)$.

8. Let $f : X \to Y$ be a projective morphism, let $\mathcal{F}$ be a coherent sheaf on X which is flat over Y, and assume that $\mathrm{H}^i(X_y, \mathcal{F}_y) = 0$ for some i and some $y \in Y$. Then show that $\mathrm{R}^i f_*(\mathcal{F})$ is 0 in a neighborhood of y.

Proof. Since $\mathrm{R}^i f_*(\mathcal{F})$ is a coherent sheaf on Y (8.8b), it suffices to show $\mathrm{R}^i f_*(\mathcal{F})_y = 0$. The question is local on Y, so we reduce to the case Y is affine, equal to the spectrum of the local Noetherian ring $A = \mathcal{O}_y$, and y is the closed point of Y. Let $\hat{A}$ denote the completion of A with respect to its maximal ideal $\mathfrak{m}$. Consider the natural isomorphism of (11.1)

$$\widehat{\mathrm{R}^i f_*(\mathcal{F})}_y \cong \varprojlim \mathrm{H}^i(X_n, \mathcal{F}_n).$$

We claim it suffices to show the right hand side is zero. Indeed, by (8.2) and (8.5), we have the isomorphism

$$\mathrm{R}^i f_*(\mathcal{F})_y \cong \mathrm{H}^i(X, \mathcal{F}),$$

and according to (A.M., 10.13), we also have the isomorphism

$$\widehat{\mathrm{R}^i f_*(\mathcal{F})}_y \cong \mathrm{H}^i(X, \mathcal{F}) \otimes \hat{A},$$

so the claim follows from the fact that $\hat{A}$ is a fathfully flat A-algebra.

Now, we will show the natural map $\mathrm{H}^i(X_{n+1}) \to \mathrm{H}^i(X_n, \mathcal{F}_n)$ is injective for every $n > 1$. Admitting this for the moment, since $X_1 = X_y$ and $\mathcal{F}_1 = \mathcal{F}_y$, it follows by induction that

$$\mathrm{H}^i(X_n, \mathcal{F}_n) \hookrightarrow \mathrm{H}^i(X_y, \mathcal{F}_y) = 0,$$

which in turn implies the inverse limit is zero, which is what we want.

Consider the exact sequence of A-modules

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^n \longrightarrow 0.$$

Since $\mathcal{F}$ is flat over Y, tensoring with $\mathcal{F}$ is an exact functor, so we have an exact sequence

$$0 \longrightarrow \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow \mathcal{F}_{n+1} \longrightarrow \mathcal{F}_n \longrightarrow 0.$$

Taking the long exact sequence in cohomology gives

$$\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \longrightarrow \mathrm{H}^i(X_{n+1}, \mathcal{F}_{n+1}) \longrightarrow \mathrm{H}^i(X_n, \mathcal{F}_n),$$

so it suffices to show the term on the left is zero. Since $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is a finite-dimensional vector space over $k(y)$, and cohomology commutes with direct sums, we have the isomorphism

$$\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \cong \mathrm{H}^i(X, \mathcal{F} \otimes k(y)) \otimes_{k(y)} \mathfrak{m}^n/\mathfrak{m}^{n+1},$$

where $k(y)$ is the constant sheaf $k(y)$ on the closed point y of Y. Thus, we can apply the isomorphism of (9.4)

$$\mathrm{H}^i(X, \mathcal{F} \otimes k(y)) \cong \mathrm{H}^i(X_y, \mathcal{F}_y) = 0$$

to conclude that $\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) = 0$.

□